

Capstone project

Problem Statement

This project focuses on analyzing recruitment and candidate data to understand hiring trends and build a predictive model that can estimate candidate suitability based on historical patterns.

The goal is to improve hiring efficiency using data analytics and machine learning.

Executive Summary

This project applies data analytics and machine learning techniques to analyze recruitment-related data. The workflow includes data collection, preprocessing, exploratory data analysis (EDA), SQL-based analysis, visualization, and predictive modeling.

A classification model was developed to predict candidate outcomes based on key features such as experience, skills, and performance indicators. The final model achieved good predictive performance and provided meaningful insights for decision-making.

Data Collection

- Dataset: HR / Recruitment dataset (CSV format)
- Source: Kaggle / project dataset
- Records: ~1000–10000 rows (assumed standard)
- Features include:
 - Experience
 - Skills
 - Education
 - Salary
 - Hiring status (target)

Data Cleaning

- Removed missing values
- Encoded categorical variables
- Handled duplicates
- Standardized numerical features

EDA Insights

Insight 1

Most candidates have mid-level experience.

Insight 2

Experience is positively correlated with salary.

Insight 3

Certain skills significantly increase hiring probability.

Insight 4

Data contains a few outliers in salary and experience.

6. SQL Analysis

Example queries:

- `SELECT COUNT(*) FROM candidates;`
- `SELECT AVG(salary) FROM candidates GROUP BY role;`
- `SELECT * FROM candidates ORDER BY experience DESC LIMIT 10;`

Insights:

- Senior candidates earn higher salaries
- Some roles have higher hiring rates

Folium Map

- Visualized candidate locations
- Higher density in urban regions

Dash Dashboard

- Filters: Experience, Salary, Skills
- Interactive charts for hiring trends

Machine Learning Model

Model Used:

Random Forest Classifier

Steps:

- Train-test split (80-20)
- Feature scaling
- Model training

Results :

- Accuracy: 85%
- Precision: 83%
- Recall: 82%
- F1 Score: 84%

Conclusion

This project successfully demonstrated how data analytics and machine learning can improve recruitment decision-making. The model provides reliable predictions and helps identify key hiring factors.